Lecture Notes
EC226 Term 1 2025/26

Misspecification

Readings:

- Stock and Watson (2003): Chapters 7.5 + 9.2
- Dougherty (2016): Chapter 6
- Wooldridge (2013): Chapter 9

Contents

1 Introduction

In this section we will consider 2 types of misspecification

1. omission of relevant variables

2. inclusion of irrelevant variables.

2 Omission of relevant variables

2.1 Properties of estimators

Consider the case where we estimate the false model, equation (F1), as opposed to the true model, equation (T1):

$$y_i = \delta_0 + \delta_1 x_i + \eta_i \tag{F1}$$

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \varepsilon_i \tag{T1}$$

and the error ε_i satisfies the CLRM assumptions and in equation (F1) we have $\eta_i - \bar{\eta} = \beta_2(z_i - \bar{z}) + \varepsilon_i$. Estimating equation (F1) (which has z_i as an omitted variable) by OLS yields

$$\hat{\delta}_1 = \beta_1 + \frac{\sum_i (x_i - \bar{x})(\eta_i - \bar{\eta})}{\sum_i (x_i - \bar{x})^2} \tag{1}$$

Substituting out $\eta_i - \bar{\eta}$ we get:

$$\begin{aligned}
\hat{\delta}_1 =& \beta_1 + \frac{\sum_i (x_i - \bar{x})[\beta_2(z_i - \bar{z}) + \varepsilon_i]}{\sum_i (x_i - \bar{x})^2} \\
=& \beta_1 + \beta_2 \frac{\sum_i (x_i - \bar{x})(z_i - \bar{z})}{\sum_i (x_i - \bar{x})^2} + \frac{\sum_i (x_i - \bar{x})\varepsilon_i}{\sum_i (x_i - \bar{x})^2}
\end{aligned}$$

In expectation the final term must be zero as $E(\varepsilon_i | x_i) = 0$, we get:

$$E(\hat{\delta}_1) = \beta_1 + \beta_2 \frac{cov(x_i, z_i)}{var(x_i)} \neq \beta_1 \tag{2}$$

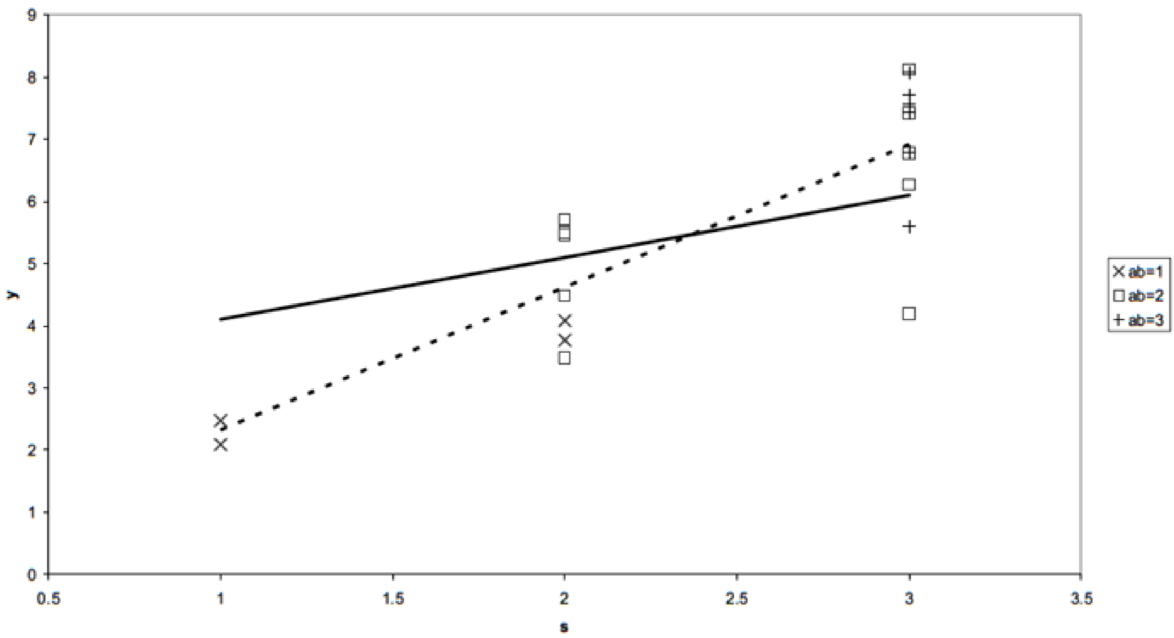
OLS is therefore biased, unless

1. $\beta_2 = 0$ in which case z_i is not omitted from the (F1) equation.

2. $cov(x_i, z_i) = 0$, that is, the omitted variable z_i is unrelated to the variable x_i.

The Bias $= \beta_2 \frac{cov(x_i, z_i)}{var(x_i)}$ (from (2).
If the coefficients are biased then the standard errors (and hence t-ratios) are wrong and all hypothesis testing is wrong. This idea can be generalised to many explanatory variables. In general omitting relevant variables will yield biased coefficients, unless the omitted variable(s) are unrelated to ALL of the included explanatory variables. Figure 1 show the bias in estimating a wage equation where y =wage, x= education and ability $z = ab$ is an omitted variable for a world in which there are 3 levels of ability: low ($ab = 1$), medium ($ab = 2$) and high ($ab = 3$).

Figure 1: Scatter plot of earnings versus schooling



2.2 Detecting omitted relevant variable

Consider the case when a relevant variable is omitted from the equation, such that the true model includes z_i and the false model has excluded this variable:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 z_i + \varepsilon_i \tag{T2}$$

$$y_i = \delta_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \eta_i \text{ where } \eta_i = \varepsilon_i + \beta_3 z_i \tag{F2}$$

Clearly $E(\eta_i) = E(\varepsilon_i) + \beta_3 z_i = \beta_3 z_i \neq 0$ and the error term, η_i, contains the influence of z_i. In which case a regression of the form:

$$\eta_i = \gamma_0 + \gamma_1 z_i + \varepsilon_i$$

would have a significant term for γ_1. However, η_i is unobserved, although we can use the residuals from (F2), denoted, $\hat{\eta}_i$, as a proxy for these and the regression is

$$\hat{\eta}_i = \gamma_0 + \gamma_1 z_i + \varepsilon_i \tag{3}$$

and the null and alternative hypotheses are: $H_0 : \gamma_1 = 0$ (z_i is not omitted from the equation) $H_1 : \gamma_1 \neq 0$ (z_i is important in determining the residuals - therefore cannot be random).
However, z_i is not known and therefore a test for omitted relevant variables does not really exist.

2.3 Detecting incorrect functional form

There is a test for incorrect functional form. Suppose there is a true nonlinear relationship between y and x, but you incorrectly estimate a linear model for between y and x,

$$y_i = f(x_i; \beta) + \varepsilon_i \tag{T3}$$

$$y_i = \delta_0 + \delta_1 x_i + \eta_i \tag{F3}$$

Digression: Taylor Series Expansion

Consider some general function of z, which we can denote as $\phi(z)$. This can always be approximated as an n^{th} ordered polynomial plus a remainder term.

$$\phi(z) = \left[\frac{\phi(z_0)}{0!} + \frac{\phi'(z_0)}{1!}(z - z_0) + \frac{\phi''(z_0)}{2!}(z - z_0)^2 + \ldots + \frac{\phi^n(z_0)}{n!}(z - z_0)^n\right] + R_n$$

For example if: $\phi(z) = \exp(z)$, then $\phi'(z) = \exp(z)$, $\phi''(z) = \exp(z)$, etc if in addition we take $z_0 = 0$ in which case

$$\begin{aligned}\exp(z) &\approx \frac{\exp(0)}{0!} + \frac{\exp(0)}{1!}(z-0) + \frac{\exp(0)}{2!}(z-0)^2 + \frac{\exp(0)}{3!}(z-0)^3 + \frac{\exp(0)}{4!}(z-0)^4 + R_4 \\ &\approx 1 + z + \frac{1}{2}z^2 + \frac{1}{6}z^3 + \frac{1}{24}z^4 + R_4\end{aligned}$$

For example, suppose we have as the true (T3) equation

$$y_i = f(x_i; \beta) + \varepsilon_i = \beta_0 + \exp(\beta_1 x_i) + \varepsilon_i$$

Define $z = \beta_1 x$ and then we have $\phi(z) = \exp(z)$ and from the digression above we know that a 2nd order Taylor series expansion will yield:

$$\phi(z) = 1 + z + \frac{z^2}{2} + r$$

In which case

$$exp(\beta_1 x) \approx 1 + \beta_1 x + \underbrace{\frac{\beta_1^2 x * 2}{2}}_{\beta_2 x^2} + r$$

Substituting this into the (T3) equation we have:

$$y_i = (\beta_0 + 1) + \beta_1 x_i + \beta_2 {x_i}^2 + w_i \tag{T4}$$

where $w_i = \varepsilon_i + r$. The (F3) equation above therefore has an omitted relevant variable (the quadratic term) and hence the error term in the (F3) equation is:

$$\eta_i = \beta_2 {x_i}^2 + w_i$$

a regression of the residuals from the F3 equation $\hat{\eta}_i$ on powers of x_i therefore aims at picking up potential incorrect functional form in the equation.

To proxy incorrect functional form in the multivariate equation

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots + \beta_k x_{ki} + \varepsilon_i \tag{4}$$

The test for incorrect functional form (the simplest version of the RESET(2) test) uses $z_i = \hat{y}_i^2$ in equation (4), the estimated model is then written as:

$$\hat{\varepsilon}_i = \delta_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \ldots + \delta_k x_{ki} + \gamma_1 \hat{y}_i^2 + v_i \tag{5}$$

where $x_{1i}, x_{2i}, \ldots, x_{ki}$ are the explanatory variable(s) in equation (4) and are included to ensure the degrees of freedom are correct and the RESET test is then an F-test

$$F = \frac{(RSS^R - RSS^U)/d}{RSS^u/DoF}$$

where $RSS^R = RSS^{(4)}$, $RSS^U = RSS^{(5)}$, d=1 and $DoF = n-(k+2)$. Other versions of the RESET test would add $\hat{y}_i^3$ and $\hat{y}_i^4$ into equation 5 and then test the joint significance of the 3 fitted value terms.

NOTE:

As $\hat{y}_i^2$ is only a proxy for the potentially omitted variables, this test has very **low power**, implying that if you fail this test you have problems of misspecification/ incorrect function form, and if you pass the test you might still have problems of misspecification/incorrect functional form.

In Stata the RESET test can be undertaken after estimation by typing:

estat ovtest → Stata includes not only the squared term, but also the cubed term and the quartic term in the RESET test (see Appendix A for an example of misspecified equations).

3 Inclusion of irrelevant variables

3.1 Properties of estimators

The true model now excludes the z_i term and the false model includes it

$$y_i = \beta_0 + \beta_1 x_i + u_i \tag{T5}$$

$$y_i = \delta_0 + \delta_1 x_i + \delta_2 z_i + \nu_i \tag{F5}$$

where $\delta_2 = 0$. In this case $E(\hat{\delta}_1) = \beta_1$, that is, the OLS estimate of the estimated model is unbiased; however, the standard errors are generally too large compared with those obtained from the true model (T5), this implies that the t-ratios will be too small, implying that some coefficients might be found to be insignificant, which are truly significant.

This can be seen using the idea of partitioned regression because in the (T5) equation

$$V(b_1) = \frac{\sigma^2}{\sum\limits_{i=1} (x_{1i} - \bar{x}_1)^2}$$

where $\sum\limits_{i=1} (x_{1i} - \bar{x}_1)^2$ is the total amount of variation in x_1 whereas in the (F5) equation

$$V(\hat{\delta}_1) = \frac{\sigma^2}{\sum\limits_{i=1} (\tilde{x}_{1i} - \bar{\tilde{x}}_1)^2}$$

where $\sum\limits_{i=1} (\tilde{x}_{1i} - \bar{\tilde{x}}_1)^2$ is the variation in x_1 over and above z.

Now we know

$$(1 - R^2_{x_1}) \sum_{i=1} (x_{1i} - \bar{x}_1)^2 = \sum_{i=1} (\tilde{x}_{1i} - \bar{\tilde{x}}_1)^2$$

where $R^2_{x_1}$ is the R-squared from a regression of x_1 on z and so

$$\sum_{i=1} (x_{1i} - \bar{x}_1)^2 \geq \sum_{i=1} (\tilde{x}_{1i} - \bar{\tilde{x}}_1)^2$$

If $R^2_{x_1} = 0$ then $\sum\limits_{i=1} (x_{1i} - \bar{x}_1)^2 = \sum\limits_{i=1} (\tilde{x}_{1i} - \bar{\tilde{x}}_1)^2$ and there is no cost to estimating the (F5) equation.

3.2 Detecting inclusion of irrelevant variables

This can simply be undertaken by undertaking a t-test of the individual coefficients.

Appendices

A Exercise of omitted relevant variable

1 The demand for a good is believed to by determined as:

$$q_i = \delta_0 + \delta_1 p_i + \eta_i \quad i = 1990 \ldots 2005$$

(1) Where q is quantity and p is the price of the good. You know that over the period of 16 year price, p, has been rising. In actual fact equation (1) has an omitted relevant variable, the price of the other good, po. Work out the direction of the bias from estimating equation (1) by OLS if:

(a) The other good is a complement and its price has been falling.

(b) The other good is a complement and its price has been rising.

(c) The other good is a substitute and its price has been falling.

(d) The other good is a substitute and its price has been rising.

(e) You are now told that in estimating equation (1) there is a further omitted relevant variable, income, y. Over the time income has been rising. Under which condition (a)-(d) can you say anything about the direction of bias of the coefficient on p estimated in (1).

Answer The true equation is:

$$q_i = \beta_0 + \beta_1 p_i + \beta_2 po_i + \varepsilon_i$$

Therefore the estimated equation is biased as:

$$b_1 = \beta_1 + \beta_2 \frac{cov(p, po)}{var(p)}$$

(a) Complement implies: $\beta_2 < 0,$ Falling price implies: $cov(p, po) < 0 \Rightarrow$ Bias > 0

(b) Complement implies: $\beta_2 < 0,$ Rising price implies: $cov(p, po) > 0 \Rightarrow$ Bias < 0

(c) Substitute implies: $\beta_2 > 0,$ Falling price implies: $cov(p, po) < 0 \Rightarrow$ Bias < 0

(d) Substitute implies: $\beta_2 > 0,$ Rising price implies: $cov(p, po) > 0 \Rightarrow$ Bias > 0

(e) The true equation is:

$$q_i = \alpha + \beta_1 p_i + \beta_2 po_i + \beta_3 y_i + \varepsilon_i$$

Therefore the estimated equation is biased as:

$$b_1 = \beta_1 + \beta_2 \frac{cov(p, po)}{var(p)} + \beta_3 \frac{cov(p, y)}{var(p)}$$

Normal good $\beta_3 > 0,$ Incoming rising implies: $cov(p, y) > 0 \Rightarrow \beta_3 \frac{cov(p,y)}{var(p)} > 0$ so can only sign overall bias in cases (a) and (d).

B Stata Commands of Reset Test

```
/* Running the main regression and saving residuals and fitted values */
reg lhourpay age c.age#c.age edage i.sex i.ethnic if(edage>=0 & edage<=95 & ethnic~=.)
predict res1, residual
predict yhat, xb
/* Conducting the Stata Reset test and saving F-value */
estat ovtest
scalar rFa=r(F)
/* Constructing the RESET test by hand lhourpay dependent variable */
reg lhourpay age c.age#c.age edage i.sex i.ethnic c.yhat#c.yhat c.yhat#c.yhat#c.yhat /*
*/ c.yhat#c.yhat#c.yhat#c.yhat  if(edage>=0 & edage<=95 & ethnic~=.)
test c.yhat#c.yhat c.yhat#c.yhat#c.yhat c.yhat#c.yhat#c.yhat#c.yhat
scalar rFb=r(F)
/* Constructing the RESET test by hand residuals dependent variable */
reg res1 age c.age#c.age edage i.sex i.ethnic c.yhat#c.yhat c.yhat#c.yhat#c.yhat /*
*/ c.yhat#c.yhat#c.yhat#c.yhat  if(edage>=0 & edage<=95 & ethnic~=.)
test c.yhat#c.yhat c.yhat#c.yhat#c.yhat c.yhat#c.yhat#c.yhat#c.yhat
scalar rFc=r(F)
/* Comparing the three alternative tests */
scalar list rFa rFb rFc
```

C R Commands of Reset Test

```
#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.
#I used the following libraries over Term 1:

library(tidyverse)
library(reshape2)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(Hmisc)
library(ggpubr)
library(rstatix)
library(groupedstats)
library(car)
library(lmtest)
library(olsrr)
library(estimatr)
library(tseries)
library(margins)
library(ivreg)

#Restricting the sample and the number of variables
qlfs18_1 <- qlfs18 %>%
  filter(!is.na(lhourpay) & !is.na(ethnic) & edage1>=0 & edage1<=95) %>%
  select(lhourpay, age, edage1, sex, ethnic)
# Estimating basic model
model20 <-lm(lhourpay ~ age + I(age^2) + edage1 + sex + ethnic,
          data = subset(qlfs18_1, edage1>=0 & edage1<=95))
qlfs18_1$res20 <- resid(model20)
qlfs18_1$yhat20 <- fitted(model20)
reset_testa <- resettest(lhourpay ~ age + I(age^2) + edage1 + sex + ethnic,
               power = 2:4, type="fitted", data = subset(qlfs18_1, edage1>=0 & edage1<=95))
fstat_reseta <- reset_testa$statistic
model_resetb <- lm(lhourpay ~ age + I(age^2) + edage1 + sex + ethnic +
                I(yhat20^2) + I(yhat20^3) + I(yhat20^4),
                data = subset(qlfs18_1, edage1>=0 & edage1<=95))
reset_testb <- linearHypothesis(model_resetb, c("I(yhat20^2)=0",  "I(yhat20^3)=0",
                "I(yhat20^4)=0"), test="F")
fstat_resetb <- reset_testb$F[2]
model_resetc <- lm(resid(model20) ~ age + I(age^2) + edage1 + sex + ethnic
                + I(yhat20^2) + I(yhat20^3) + I(yhat20^4),
                data = subset(qlfs18_1, edage1>=0 & edage1<=95))
```

```
reset_testc <- linearHypothesis(model_resetc, c("I(yhat20^2)=0",  "I(yhat20^3)=0",
                  "I(yhat20^4)=0"), test="F")
fstat_resetc <- reset_testc$F[2]
c(fstat_reseta, fstat_resetb, fstat_resetc)
```

References

Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.

Stock, J. and Watson, M. W. (2003). *Introduction to Econometrics*. Prentice Hall, New York.

Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.